244 Basic Idea of Question Answering System with Chattering Functions

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ABSTRACT

Nowadays, applications for computers are widespread in the field of natural language question answering (QA) systems in order to achieve the goal of making the human–computer interaction more like human–human interaction. In this paper, we present the basic idea of a QA system that takes advantage of *chattering functions* in order to choose the most suitable answer for a given user's question and make the user-system interaction more like human-human.

1. Introduction

Recently the wealth of information on the Internet makes it an attractive and useful resource as a knowledge base for QA systems. Although the huge amount of information that can be retrieved from the Web, presented in heterogeneous formats, makes it difficult to extract the correct answer to a given question.

Lots of research is being carried out around QA systems in order to achieve the goal of taking a natural language question and return the correct answer as a short natural language response. Systems like MULDER[1] and START[2] have made an effort to solve this problem but the correct answer selection still remains a difficult task to solve.

As an endeavor to make more suitable the selection process of a correct answer for an user's question, we propose a QA system with ***chattering functions*** based on the Word Wide Web (WWW). By means of the chattering functions, we can obtain more precise information from the user about the expected answer, allowing the system to select the most appropriate response for the user's request, and we can reach, at the same time, a human-computer interaction more like human-human one. In case that the speech recognition process fails, the system can initiate a repair chat with the user in order to clarify the user's expression.

In this paper, we emphasize the Dialogue Mode process in which the interaction with the user serves as a means for making easier the answer selection.

The development of an experimental system and experiments for performance evaluation are left as a future work.

2. Dialogue Mode.

2.1 Basic Idea.

The process of our proposed method is shown in Fig. 1.

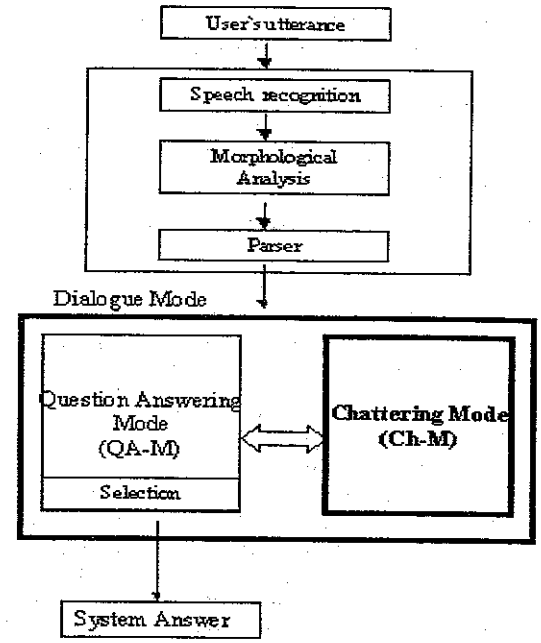


Fig. 1. Process Flow Chart

The user's utterance is changed into text in the speech recognition step. Then morphological analysis and parse process are carried out. In the Dialogue Mode process, the system analyzes the user's utterance and carries out a chat in order to select the best answer to the user's question. Fig. 2 shows a more detailed view of the Dialogue Mode process. In the Dialogue Mode process, the Question Answering Mode process (QA-M) and Chattering Mode process (Ch-M) are performed. As a result the system answer is obtained.

2.2 Question Answering Mode Process (QA-M).

During this process the user's question is analyzed. In the query formation step, non-content words (like question words, auxiliary verbs, etc) are cut off and the remaining words (nouns, verbs) are expanded with synonyms and morphological forms in order to maximize the recall of retrieved documents. For accomplishing this task, the WordNet[3] dictionary is used. The composed queries are sent to the search engine and documents are obtained from the Web.

In the information extraction process more relevant answers are selected according to the user's information obtained by the Chattering Mode (Ch-M) process. Then the answer selection process is carried out and the best answer is selected as the one of the highest frequency within the ones obtained by the information extraction process.

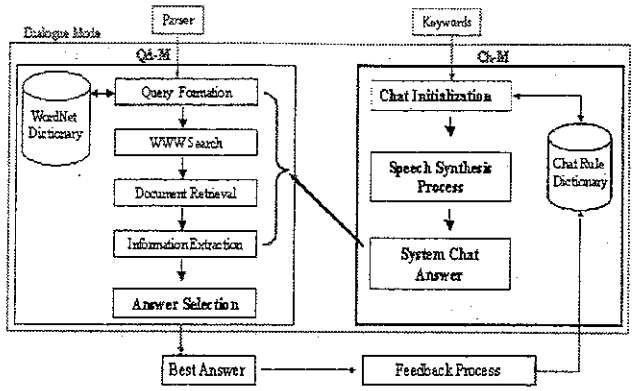


Fig. 2. Dialogue Mode

For example, the user's question: "What is the highest mountain?" Let us suppose that the user wants to know what is the highest mountain in the "Caribbean". But, for some reason, she/he does not give specific information to the system. A normal QA system will return the answer "Everest" as the correct answer, which is true but it is not the most suitable answer for this user. In those cases, our system works trying to obtain useful information from the user in order to give a suitable answer. The system works like follows: after the speech recognition process (in the case that it did not fail), morphological analysis and parser process, the QA-M process the result cutting off "what" and "the" as non-content words and "mountain (noun)" is set as a keyword. Then the content words are expanded using WordNet. We get queries like:
[highest <text> mountain <text> is]
[highest<text>peak<text>is]
[highest <text> mountain <text> was]
and so on.

Then we send those queries to the search engine and obtain the document retrieval. In the information extraction process, keywords from the user's response during the Ch-M process are added in order to improve the answer selection according with the user's expectations.

2.3 Chattering Mode Process (Ch-M).

In case that the user's utterance could not be "well understood" by the system (speech recognition fails), this Ch-M starts a *repair chat* with the user using expressions like "Could you repeat that", "Sorry I didn't get it", "Try using another expression", in order to keep the interaction with the user alive in a natural way.

Otherwise, this process starts being carried out at the same time that the QA-M starts. The goal of this process is to obtain from the user relevant information about the user's request and make the user-computer interaction more like human-human interaction.

The chat initialization step uses keywords (nouns, obtained from the parser process) for starting a chat with the user about his/her question. This is an ELIZA[4]-like chat system.

In this step the chat rule dictionary is used. On it, rules about the way of asking for information to the user are stored. For example,
"So, you're interested in @0, why?"
"I see. What places do you prefer for @1?"
"So, you like @2 places, could you be more specific?
and so on. Where @0, @1 are variables (noun keywords: @0 from the user's original question, @1 from the user's answer to the system question). @2 is an adjective keyword. For our example, we get then, system interaction like:

"So, you're interested in mountain, why?" A possible user's answer could be: "I like to climb! I like adventure!" From this the new keyword "adventure (noun)" is selected and sent to the QA-M, which adds this new keyword to the former queries and searches again on the Web. This process is repeated once again (we set a searching loop of 3 times maximum) and the system can use "I see. What places do you prefer for adventure?" since the last keyword was a noun. In order to answer this question the user has to be specific about his/her preferences, then, an answer could be: "I like places like Caribbean or South America". Becoming those two proper names the new keywords that are sent to the QA-M, and being added to the former queries the system obtains a list of possible answers, according to the user's expectations. The best answer is selected as the one of the highest frequency among the retrieved documents. For our example:
"The Dominican Republic is home to the Caribbean's highest peak, Pico Duarte, (10,128ft)"
"Highest mountain in South America, Aconcagua, Argentina 22,834ft", is the most suitable answer for the user instead of being "The highest peak: Everest, Himalayas, 29,035ft".

3. Feedback Process

During this the chat rule dictionary is updated. The user judges whether the given answer satisfied his/her requirements or not. If the system's answer is judged as correct then the used rules improve their value, otherwise, their value decrease.

4. Conclusion and Future Work

We have presented the basic idea of an approach to QA system that takes advantage of chattering functions in order to make the answer selection process more suitable to the user's requirements and achieve a more like human-human interaction between the user and the computer.

Several activities are planned in the near future. Experiments and comparison of the performance of the system with and without the Chattering Mode are some of the main points for examination.

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